

- 4 White-clawed crayfish, *Austropotamobius pallipes*, live in rivers and lakes in Europe. In the 1850s, the North American signal crayfish, *Pacifastacus leniusculus*, was introduced to Europe. The introduced species carried a pathogen that causes a disease known as crayfish plague. This disease kills *A. pallipes*. Since 1850, the population size of *A. pallipes* has reduced in many areas of Europe due to the spread of crayfish plague.

- (a) North American *P. leniusculus* can carry the crayfish plague pathogen without showing symptoms because they have evolved resistance to it.

Explain how *P. leniusculus* could have evolved resistance to the crayfish plague pathogen.

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- (b) It is difficult to locate crayfish because they live underwater. After an outbreak of crayfish plague in one country in Europe, researchers used an environmental DNA technique to find the locations in a river where populations of *A. pallipes* were still surviving. Water samples were taken from locations along the river. Each water sample was filtered to obtain any cells or DNA that had been released by organisms into the water. The polymerase chain reaction (PCR) was carried out on this DNA using primers specific to *A. pallipes* DNA.

- (i) Outline **and** explain the steps that occur in PCR.

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- (ii) Suggest how the researchers were able to discover suitable sequences for the primers.

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- (c) The PCR technique used in this research is quantitative. A fluorescent dye binds to the DNA and the fluorescence is monitored throughout the process, showing the quantity of DNA present.

Fig. 4.1 shows standard curves for four known concentrations of DNA, **A**, **B**, **C** and **D**, and the result obtained for *A. pallipes* DNA at one location along the river (dotted line).

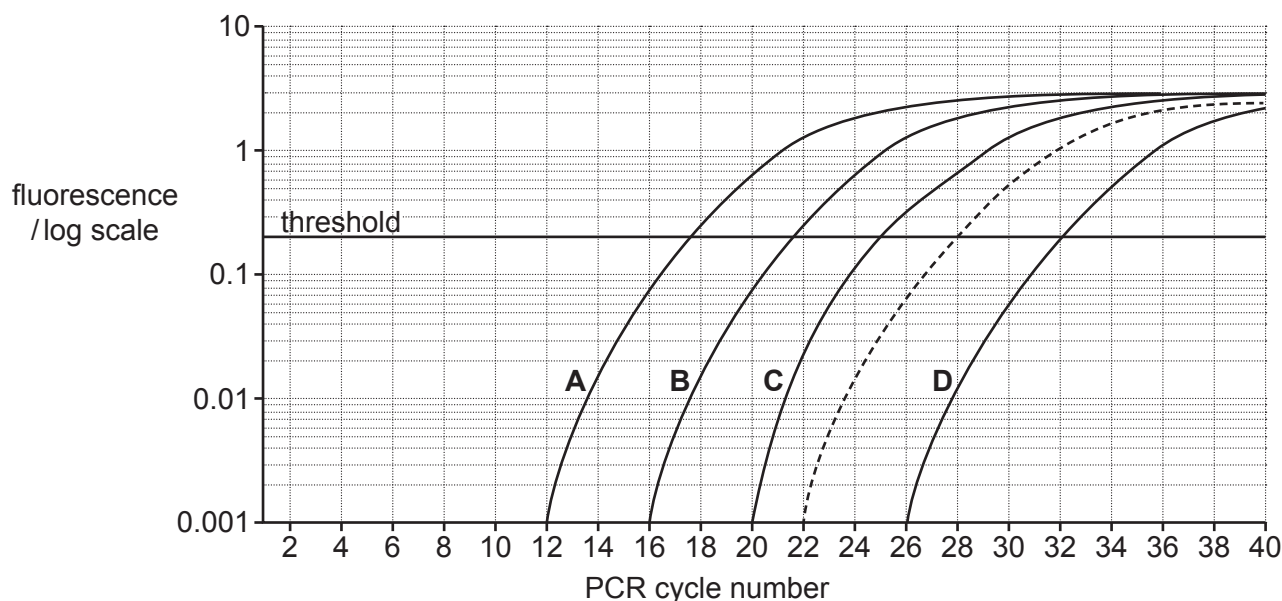


Fig. 4.1

- (i) State the letter of the curve with the highest starting concentration of DNA.

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- (ii) The quantity of *A. pallipes* DNA reaches a threshold (marked as a horizontal line on Fig. 4.1) at 28 cycles. The DNA in curve **C** reaches the threshold after 25 cycles. In one cycle of PCR the concentration of DNA doubles.

Calculate the relative difference in the starting concentrations of *A. pallipes* DNA and the DNA in curve **C**.

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- (d) Fluorescent molecules have other uses in gene technology.

Fig. 4.2 shows an agar plate with normal bacteria and genetically modified bacteria. The genetically modified bacteria make a green fluorescent protein.

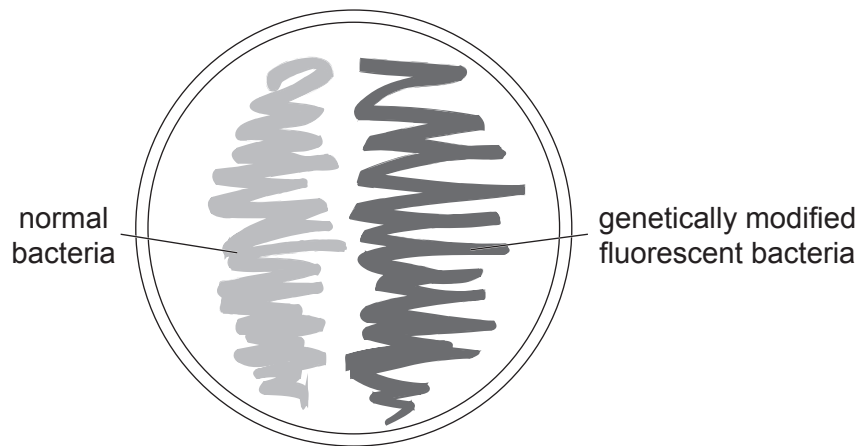


Fig. 4.2

- (i) Identify the component that is taken up by normal bacteria to produce the genetically modified bacteria in Fig. 4.2 **and** state what this component contains to allow the bacteria to make green fluorescent protein.

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- (ii) Suggest why the gene for green fluorescent protein is sometimes transferred in addition to the desired gene in genetic engineering.

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[Total: 15]